Keqi Ni

Mechanical Engineering Undergraduate | Embodied AI and Memory-Based Reinforcement Learning

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EDUCATION

Pittsburgh Institute, Sichuan University | Sep. 2023 - Jun. 2027 (expected)
B.Eng. Candidate in Mechanical Design, Manufacturing and Automation (English-taught)

GPA: **3.83/4.00**; weighted average: 89.66/100; academic rank: 6/50.
SCUPI Dean's List Scholarship (2024). IELTS: **7.0**.

RESEARCH INTERESTS

Meta-learning exploration patterns that help agents learn world dynamics and adapt to new situations or tasks; LLM-guided policy learning with memory; memory-based RL for dynamics adaptation and learning from mistakes.

RESEARCH PROFILE

Deeply motivated by fundamental problems in embodied AI, especially memory-based reinforcement learning and autonomous learning through LLM guidance. Able to digest research papers quickly, identify their core ideas, and build on them through implementation and controlled experiments. Good command of English.

RESEARCH EXPERIENCE

The Chinese University of Hong Kong | Remote Research Intern | Sep. 2026 - Nov. 2026
Supervisors: Prof. Zhongyu Li & postdoc Zike Yan

- Investigated memory-based RL and test-time training for the GMP T4 iterative pushing task, including adaptation across repeated attempts and generalization to unseen friction.
- Studied sparse terminal-return learning and joint prediction/value objectives; evaluated ID and OOD behavior with preserved causal memory and matched physical resets.

Pittsburgh Institute, Sichuan University | Research Assistant | 2025 - Present
Supervisor: Prof. Hao Qin

- Designed evaluation protocols and completed training, comparative experiments, and ablations for physics-informed high-frequency tunnel propagation modeling (IEEE T-MTT, second author).
- Proposed the spatio-sequential recurrent architecture and convolutional gating mechanism for 3-D tunnel propagation modeling (IEEE AWPL, third author).
- Explored radar-IMU gesture recognition using cross-modal JEPA representations, EMA targets, and frozen temporal-classification probes.

College of Electronics and Information Engineering, Sichuan University | 2025
Undergraduate Research Project | Supervisor: Prof. Xinzhi Zhou

- Designed ATTRN with an MLP connection between a Transformer encoder and CNN decoder; completed experiments and wrote the Method section for boiler temperature-field reconstruction.

West China Hospital Big Data Center, Sichuan University | Research Assistant | 2023 - 2024
Supervisor: Fashu Xu

- Developed an IMU-based gait-phase classifier for a powered lower-limb exoskeleton and studied reinforcement learning for exoskeleton control.

PUBLICATIONS

1. K. Wu, **K. Ni**, H. Xu, J. Wang, H. Qin, X. Zhang, and X. Zhang. "A Generalizable Physics-Informed Long-Range Attentive Network for Radio-Wave Propagation Modeling in Tunnels." *IEEE Transactions on Microwave Theory and Techniques*, 73(12):9854-9866, 2025.

2. J. Li, J. Xue, **K. Ni**, K. Wu, H. Qin, X. Zhang, and X. Zhang. "Spatio-Sequential Recurrent Network for 3-D Tunnel Propagation Modeling." *IEEE Antennas and Wireless Propagation Letters*, accepted, 2026.

3. K. Wu, **K. Ni**, L. Chen, H. Xu, J. Wang, J. Zhou, and X. Zhou. "ATTRN: Acoustic Information Encoder and Temperature Field Reconstruction Decoder Network for Boiler Temperature Field Reconstruction." *Sensors*, 25(8):2567, 2025.

TECHNICAL SKILLS

Python / NumPy, PyTorch, C++, MATLAB, Simulink, CATIA, LaTeX; experimental analysis and reproducible model evaluation.